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PMMA/TEMPO-oxidized cellulose nanofiber nanocomposite with improved mechanical properties, high transparency and tunable birefringence

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Abstract Recently, cellulose nanofibers (CNFs) have been developed as a very popular renewable and biodegradable nanofiller material for polymer nanocomposites. However, achieving good dispersion in a polymer matrix for effective reinforcement is still a challenge because CNFs are hydrophilic, whereas most polymers are hydrophobic. In this study, we report the poly(methyl methacrylate)/2,2,6,6-tetramethylpiperidyl-1-oxyl oxidized CNFs (PMMA/TOCN) nanocomposites, which show good dispersion, improved mechanical properties, excellent transparency, as well as controllable birefringence using a simple surface-modification procedure of TOCN with amine-functionalized poly(ethylene glycol). Studies conducted using transmission electron microscopy and fourier transform infrared spectroscopy showed that TOCNs were homogenously dispersed in the PMMA matrix without aggregation due to the successful surface modification of TOCN. Moreover, the nanocomposites were highly transparent and the transmittance in the visible region was as high as approximately 90%. In addition, we firstly discovered that the birefringence of the nanocomposite could be controlled by the amount of TOCN added, even achieving zero birefringence. More importantly, the tensile strength and Young's modulus of PMMA were significantly improved with the addition of TOCN. Such well-dispersed TOCN-based nanocomposites with high transparency, controllable birefringence and enhanced mechanical properties exhibit great potential for the applications in the optical devices and in the engineering field.

Keywords Nanocomposites · PMMA · TEMPO-oxidized cellulose nanofiber · Transparency · Birefringence

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Introduction

Polymer nanocomposites, mainly comprising a polymer matrix and nanofillers, have attracted considerable interest over the last decades. In some cases, neat polymers are not suitable for specific applications and nanofillers are required to reinforce or modify the properties of the polymers to meet the requirements of the application. Based on the specified applications, different nanofillers can be incorporated with polymer matrix. For example, clay is generally utilized to improve the tensile strength, flame resistance, gas-

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barrier properties for applications in packaging and automobile sectors. Several types of nanofillers are available, including carbon-related fillers [carbon nanotubes (Jia et al. 1999; Philip et al. 2003; Clayton et al. 2005), graphene (Gonçalves et al. 2010; Kuila et al. 2011)], inorganic fillers [glass fiber (Akkapeddi 2000; Yoo et al. 2011), SiO_2 particles (Carotenuto et al. 1996; Palkovits et al. 2005)], metal & metal oxides [silver (Singh and Khanna 2007; Huang and Xu 2010), TiO_2 (Khaled et al. 2007; Džunuzovic et al. 2009)], and clays [talc (Castillo et al. 2013; Patnaik et al. 2010), montmorillonite (Gao et al. 2001; Zhu et al. 2002; Valandro et al.; 2014)]. Recently, with the increasing awareness of sustainability, cellulose nanofibers have attracted considerable attention as a new type of nanofiller due to their renewability and biodegradability.

Cellulose is the most abundant organic polymer on earth and is mainly produced by green plants and bacteria. It exists in the form of nanosized microfibrils with alternating crystalline and amorphous regions (Schroers et al. 2004). Due to the existence of multiple hydrogen bonds, cellulose microfibrils are tightly hooked together. Two types of plant cellulose nanofillers have been extensively studied to reinforce polymers: microfibrillated celluloses (MFCs) (Siró and Plackett 2010; Lavoine et al. 2012; Seydibeyolu and Oksman 2008; Nakagaito et al. 2009) and cellulose nanowhiskers (CNWs) (Eichhorn et al. 2010; Capadona et al. 2009; Oksman et al. 2006). However, because MFCs comprise bundles of microfibrils of different widths and CNWs have relatively short length and low aspect ratios (Siró and Plackett 2010; Isogai et al. 2011), their reinforcing performance is lower than expected. 2,2,6,6-Tetramethylpiperidine-1-oxyl (TEMPO)-mediated oxidation and mechanical treatment were combined to isolate TEMPO-oxidized cellulose nanofibers (TOCNs) from wood cellulose (Isogai et al. 2011). Owing to their high aspect ratios (> 100), uniform widths of 3–4 nm, and high elastic moduli of 145 GPa (Isogai et al. 2011), TOCNs show a high potential as reinforcing nanofillers for polymer nanocomposites with a better reinforcing performance than other nanocellulose materials.

Over the decades, poly(methyl methacrylate) (PMMA) has been established as a popular polymer to prepare polymer nanocomposites for specific applications, for example transportation sectors, gas sensing and so on. Different types of nanofillers have been studied to reinforce or modify the properties of PMMA, including clay (Gao et al. 2001; Zhu et al. 2002), carbon nanotube (CNT) (Jia et al. 1999; Philip et al. 2003; Clayton et al. 2005), graphene (Gonçalves et al. 2010; Kuila et al. 2011). However, to the best of our knowledge, reinforcement by these nanofillers could not simultaneously improve the mechanical properties of PMMA nanocomposites and maintain their high transparency. Due to their homogenous nanoscale sizes and strong mechanical properties, TOCNs show a high potential to achieve improved mechanical properties and high transparency at the same time. The critical challenge lies in the ability to disperse TOCNs homogenously in the PMMA matrix. Different modification methods have been explored to change the surface chemistry of cellulose and improve its dispersion in solvents and polymer matrices, including non-covalent surface modification, esterification and polymer grafting (Roy et al. 2009; Habibi 2014). Here a very simple process, i.e., surface modification of the TOCN surface with PEG-NH_2 proposed by Isogai group (Araki et al. 2001; Fujisawa et al. 2013), was introduced to achieve good dispersion of TOCNs in the PMMA matrix. We chose this process because the carboxyl group of TOCN undergoes ionic interactions with the amino group of PEG-NH_2, and PEG part of PEG-NH_2 is miscible with PMMA. Therefore, we expected that good dispersion and adhesion of TOCNs in the PMMA matrix could be achieved using this simple surface-modification process leading to high-performance polymer nanocomposite.

In this study, we report the successful preparation of PMMA/TOCN nanocomposites with improved mechanical properties and high transparency. The PMMA/TOCN nanocomposites were prepared by blending PMMA and PEG-NH_2-modified TOCN in *N*,*N*-dimethylformamide (DMF) with various weight ratios. With the surface modification, TOCN showed good dispersion and adhesion in PMMA; moreover, the PMMA/TOCN nanocomposites were highly transparent and exhibited improved mechanical properties. More importantly, TOCNs were firstly investigated for their ability to tune the birefringence of the nanocomposites, which implies a high potential for possible applications of TOCNs in optical devices.

Experimental section

Materials

TOCN water dispersion (0.28 w/v%) was provided by Prof. Isogai's group at The University of Tokyo. The preparation method is described elsewhere (Fujisawa et al. 2012). PEG-NH_2 (M_w around 1000, NOF Corporation), PMMA (M_w 74,000, PDI 1.95, Sumitomo Chemical), HCl (35 wt%, Yotuhata Pure Chemicals), and DMF (94.5–94.7 wt%, Takahashi Pure Chemical), were used as received.

Preparation of PMMA/TOCN nanocomposite

HCl (0.1 mol/L) solution was added to TOCN water dispersion (0.2 w/v%) and, then kept for 12 h to obtain the TOCN hydrogel. DMF was used for the solvent exchange of the TOCN hydrogel from water to DMF, and the process was repeated 5 times. Furthermore, PEG-NH_2 was added to the TOCN organogel in DMF and maintained for 12 h with magnetic stirring (mole ratio of PEG-NH_2 to TOCN was 1:1). PMMA solution in DMF (5 wt%) was mixed with the TOCN dispersion in DMF (weight ratios of TOCN were 1, 3, 5, and 10 wt%). The mixture solution was solution-cast on the polytetrafluoroethylene (PTFE) substrate and dried at ambient pressure at 60 °C for 8 h, and then the samples were moved to a vacuum oven and maintained at 30 °C for one week. The PMMA/TOCN nanocomposite was peeled off from the PTFE substrate and hot pressed at 170 °C, 200 MPa for 10 s, followed by quenching.

Fourier transform infrared spectroscopy (FTIR)

The PMMA/TOCN nanocomposite films after hot pressing were used to measure their FTIR (JASCO-480 Plus) spectrum under transmission mode from 400 to 4000 cm^{-1} with a 4 cm^{-1} resolution. The film thickness was about 30 μm.

Transmission electron microscopy (TEM)

TEM was applied to observe the microstructure of the PMMA/TOCN nanocomposites. The ultrathin section of the nanocomposite was about 80 nm thick, prepared by an ultramicrotome (Leica Instrument). TEM imaging was conducted on a Hitachi H-7650 zero A at an accelerating voltage of 100 keV.

Optical transmittance

PMMA/TOCN nanocomposite films containing different amounts of TOCN after hot pressing were used to measure the UV–vis light transmittance (JASCO V-7200 Spectrophotometer). The film thickness was about 100 μm. The wavelength range was from 200 to 800 nm.

Thermogravimetric analysis (TGA)

TGA measurement was conducted to investigate the thermal decomposition of PMMA/TOCN nanocomposites using a Thermo Plus (TG 8120, Rigaku) with nitrogen flow and a heating rate of 30 mL/min and 10 °C/min, respectively. An approximate sample amount of 10 mg was set in an alumina pan for the TGA measurement. The heating range was from 100 to 600 °C.

Differential scanning calorimetry (DSC)

DSC was conducted using a DSC-6200 (Seiko Instruments Inc.) to determine the glass transition temperature (T_g). The amount of PMMA/TOCN nanocomposite was about 5 mg. The temperature range was from 20 to 170 °C. The heating rate was 20 °C/min and the cooling rate was 10 °C/min.

Birefringence

The birefringence of PMMA/TOCN nanocomposite films containing different amount of TOCN after hot pressing were measured by the photo-elastic method (Home-designed equipment) during the tensile testing process. The film thickness was about 100 μm, the width was 5 mm, and the length was 10 mm. The birefringence of the films was measured at ($T_g + 10$) °C and the strain speed was 10%/s.

Tensile test

The tensile test was performed at room temperature. The dumbbell-shaped samples for the tensile test were prepared by compression molding. The gage length, width and thickness of the sample were 12, 3 mm and

100 μm, respectively. The tensile properties of the specimen were evaluated using a tensile tester (TensilonUTM-II-20, Orientec Co., Ltd.) with a crosshead speed of 2 mm/min.

Results and discussion

Morphology and structure

The TEM images of TOCNs used in the PMMA/TOCN nanocomposites are shown in Fig. 1. The TOCNs have uniform widths of around 4 nm and lengths in the range of 300 ~ 700 nm, which means they have high aspect ratios of over 100, indicating their high reinforcing ability for polymer nanocomposites.

PMMA/TOCN nanocomposites with various weight ratios were prepared by solution casting and drying of the TOCN DMF dispersion and the PMMA DMF solution, followed by a hot pressing and quenching process. The PMMA/TOCN nanocomposites, which were around 100 μm in thickness, were highly transparent even with 10 wt% TOCN (Fig. 2). In general, for polymer nanocomposites, nanofiller aggregation in the polymer matrix leads to a decrease in light transparency (Fujisawa et al. 2012), therefore, to some extent, the high optical transparency of PMMA/TOCN nanocomposite films indicates that TOCNs, which have dimensions that are much smaller than optical wavelengths, were homogenously dispersed in the PMMA matrix without significant aggregation.

To further confirm the homogenous dispersion of TOCN in the PMMA matrix, the microstructures of the nanocomposites were observed by TEM. First, the microstructures of PMMA/TOCN nanocomposites with/without surface modification of the TOCNs with PEG-NH_2 were compared, as shown in Fig. 3. In the TEM images, TOCNs appear as black filaments were TOCNs and the bright region represents PMMA. Clearly, PEG-NH_2 surface modified-TOCNs were very homogenously dispersed in the PMMA matrix for PMMA/TOCN nanocomposite, whereas without surface modification, the TOCNs were severely aggregated, indicating the significant contribution of the surface modification to the good dispersion of the TOCNs in the PMMA matrix.

In addition, we investigated the surface-modified PMMA/TOCN nanocomposites with various weight ratios (Fig. 4). The images show that the TOCNs were homogenously distributed in the PMMA matrix without severe aggregation. When the TOCN contents were 1 and 3 wt%, the TOCNs were almost individually distributed in the PMMA matrices. Even when the TOCN content was 10 wt%, homogenous dispersion of TOCNs in the PMMA matrix was observed as a result of the surface modification of TOCN by PEG-NH_2. Accordingly, to some extent, the homogenous dispersion of TOCN in PMMA matrix proved the high optical transparency of the nanocomposites shown in Fig. 2.

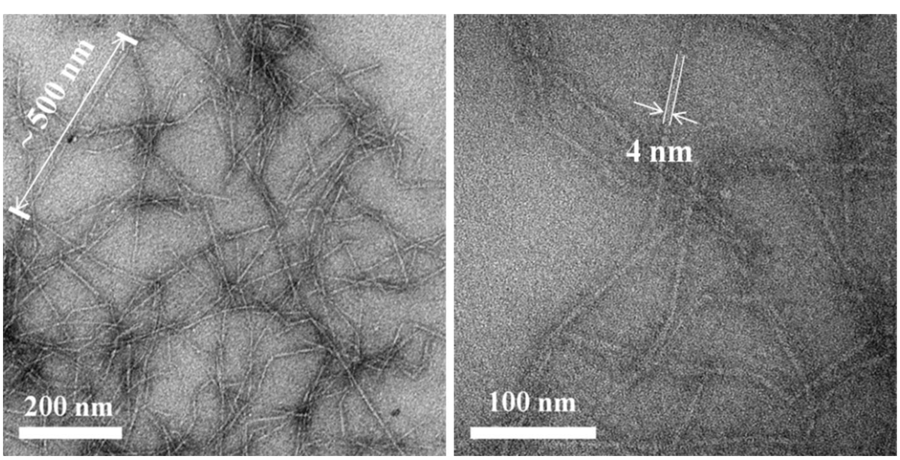

Fig. 1 TEM images of TOCNs used in PMMA/TOCN nanocomposites

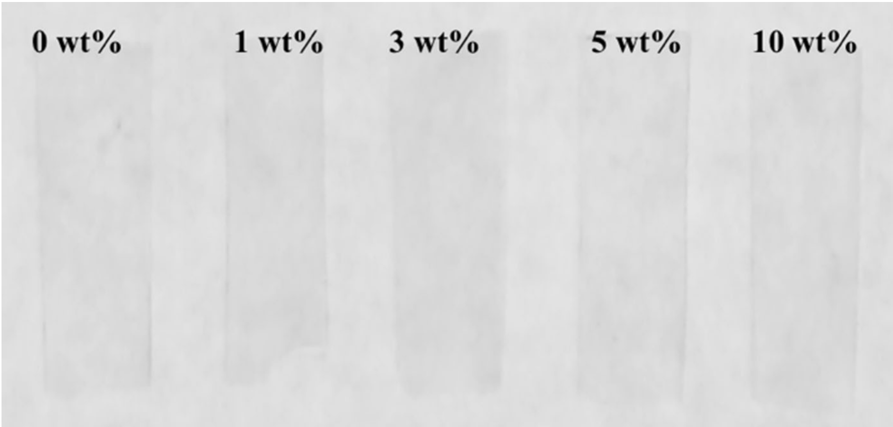

Fig. 2 Transparent PMMA/TOCN nanocomposite films with different TOCN contents

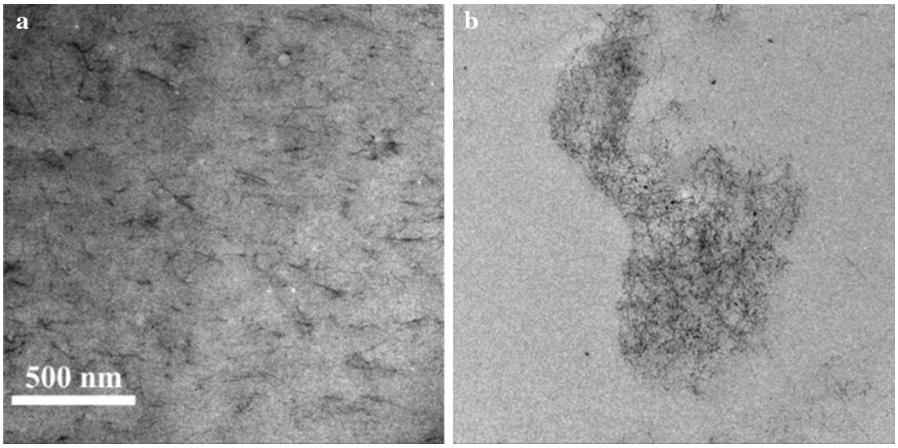

Fig. 3 TEM images of PMMA/TOCN nanocomposites (5 wt%). **a** With PEG-NH_2 surface modification. **b** Without PEG-NH_2 surface modification

FTIR was conducted to confirm the surface modification of TOCN by PEG-NH_2. Figure 5 shows the FTIR spectra of a neat TOCN-COONa film, TOCN-COOH and the PMMA/TOCN nanocomposite films with various weight ratios. The peak at 1608 cm^{-1} of neat TOCN-COONa was assigned to the C=O stretching vibration of –COONa. Moreover, this peak shifted to 1720 cm^{-1}, assigned to the C=O stretching vibration of –COOH in TOCN-COOH film after the addition of HCl. In the PMMA/TOCN nanocomposites, the peak at 1720 cm^{-1} shifted to 1681 cm^{-1} due to the surface modification by PEG-NH_2, which forming the ionic interaction between amino group of PEG-NH_2 and hydroxyl group of TOCN (Fujisawa et al. 2013), indicating the successful surface modification of TOCN with PEG-NH_2. At the same time, the peak at 1720 cm^{-1} of PMMA/TOCN nanocomposite, attributed to –COOH of TOCN, also increased with the TOCN content, which indicated that part of the carboxyl groups of TOCN did not form ionic interaction with PEG-NH_2.

Optical transmittance

As shown in Fig. 2, the PMMA/TOCN nanocomposites with TOCN contents from 1 to 10 wt% were highly transparent. In the visible light region, the transmittance of the PMMA/TOCN nanocomposite films (thickness around 100 μm) was as high as approximately 90% (Fig. 6), showing their high potential for applications in optical devices and daily-life appliances. Surprisingly, with the increase of the TOCN content, the transmittance only showed a very slight decrease, even with 10 wt% TOCN, because of the good dispersion of TOCN in PMMA matrix shown in Fig. 4, homogenous nanosize of

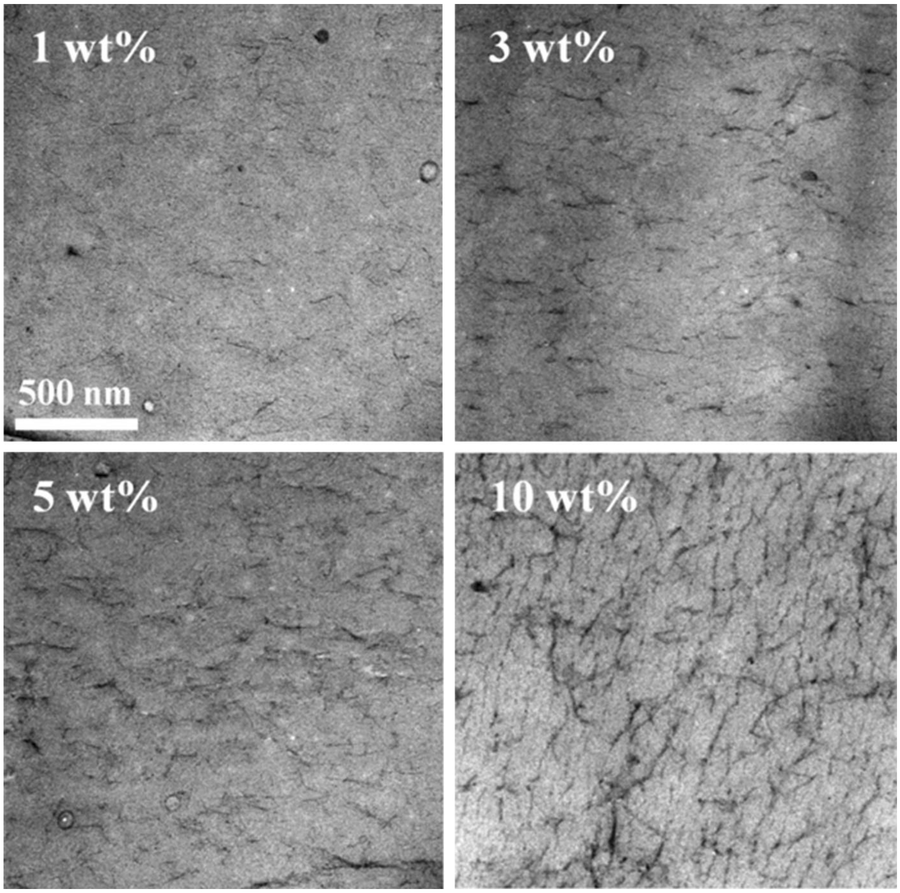

Fig. 4 TEM images of PMMA/TOCN nanocomposite films

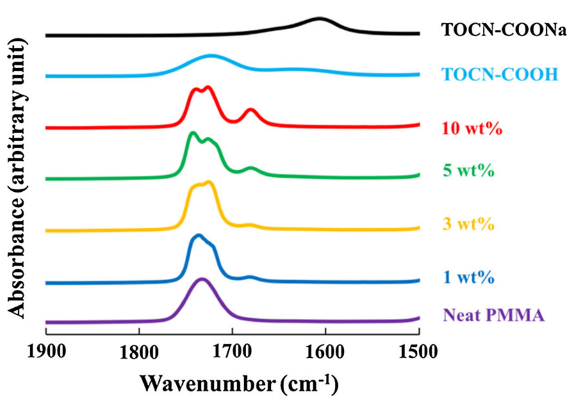

Fig. 5 FTIR spectra of TOCN-COONa, TOCN-COOH and PMMA/TOCN nanocomposite films

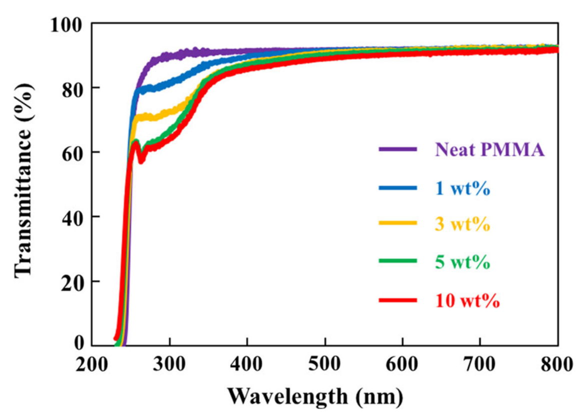

Fig. 6 Optical transmittance of the PMMA/TOCN nanocomposites

TOCN (Fig. 1) as well as no big absorption band of TOCN in the visible light region. Furthermore, by comparing the transmittances of the PMMA/TOCN nanocomposites with the various weight ratios at specific wavelengths (550 and 400 nm), we concluded that TOCN had no significant influence on the transmittance even at high loading (Fig. 7). In contrast, for the PMMA/CNT nanocomposites (Kim et al.

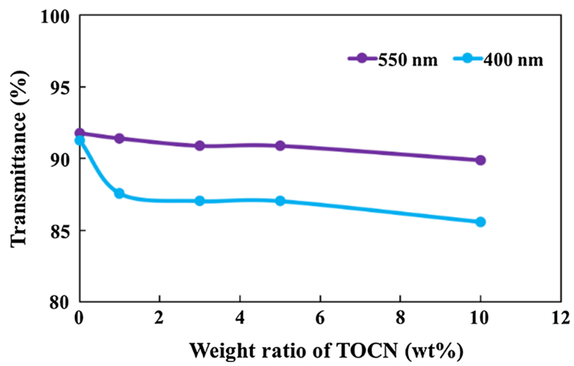

Fig. 7 Transmittance of the PMMA/TOCN nanocomposites with various weight ratios at 550 and 400 nm

2008), with only 1 wt% loading of CNT, the transmittance strongly decreased from 93% for neat PMMA to about 75% for PMMA/CNT nanocomposites. Moreover, the addition of clay also had a significant influence on the optical transparency of the PMMA/clay nanocomposites, causing a reduction in transmittance from about 82% of neat PMMA to around 50% of PMMA/clay nanocomposite with 1.46 wt% loading of clay (Kim et al. 2008). Hence, in term of transparency, TOCN is a potential candidate as a nanofiller for the preparation of high transparency-polymer nanocomposites for optical devices. Moreover, the observed absorption at around 300 nm (Fig. 6) may be due to yellowish discoloration of TOCNs during the hot pressing in sample preparation process. We supposed that there are two reasons causing the yellowish discoloration. The first reason is that interfibrillar hemiacetal linkages are formed during the hot pressing (Takaichi et al. 2014). Takaichi et al. reported that the TEMPO-oxidized wood cellulose (TOC) became yellowish and had an absorption around 300 nm after oven-dry at 105 °C for 3 h. Therefore, during the hot pressing at 170 °C for 10 s in this study, interfibrillar hemiacetal linkages would also occur. On the other hand, the second reason might be due to thermal decomposition of TOCNs during the hot pressing. Combining the facts that the pure TOCN film, which is hot pressed at the same conditions, showed absorption at around 300 nm (Supporting Information, S1) and that the following TGA results (Fig. 8) showed about 3% weight loss of TOCNs at 170 °C, the thermal degradation during the hot pressing of TOCNS in PMMA/TOCN nanocomposites caused the absorption at around 300 nm.

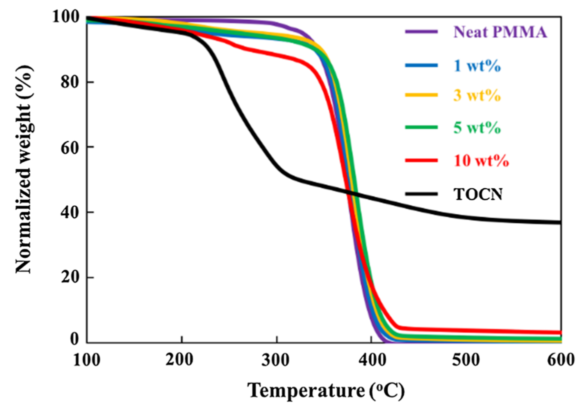

Fig. 8 TGA curves of TOCN, neat PMMA and PMMA/TOCN nanocomposites (N_2 flow of 30 mL/min)

Thermal properties

The thermogravimetric curves of neat PMMA, PMMA/TOCN nanocomposites with different TOCN contents prepared by solution blending and of the TOCN film are shown in Fig. 8. The neat TOCN tended to decompose around 150 °C, and then dramatically decomposed at 200 °C, producing bio-oil, such as levoglucosan and hydroxyactone (Shen and Gu 2009). It was reported that TOCN has a lower decomposition temperature than that of the original cellulose, because the anhydroglucuronic units in TOCN degrade at lower temperature by decarbonation of the sodium carboxylate groups, followed by the cellulose molecular structure (Fukuzumi et al. 2010). Based on the fact that TOCN started to decompose at around 150 °C and the temperature of the hot pressing was also 170 °C, TOCN slightly thermally decomposes during hot pressing process, causing the absorption at around 300 nm of the PMMA/TOCN nanocomposites, which corresponds to the analysis mentioned above.

Moreover, as shown Fig. 8, neat PMMA started to decompose at around 200 °C and ended its decomposition around 400 °C with almost no products left. In addition, for the PMMA/TOCN nanocomposites containing 1, 3, and 5 wt% of TOCN, water evaporation and slight decomposition of TOCN occurred between 100 and 200 °C, following by the thermal decomposition of TOCN and slight decomposition of PMMA over 200 °C. Decomposition of PMMA in the PMMA/TOCN nanocomposites started at around 300 °C and ended at around 400 °C. A similar decomposition tendency of PMMA/TOCN nanocomposite containing 10 wt% TOCN was observed. Notably, the T_g of neat PMMA and PMMA/TOCN nanocomposites with various TOCN weight ratios was around 100 °C (Supporting Information, Table 1).

Birefringence

Owing to their high transparency, easy processing, and light weight, optical polymers, such as PMMA are widely applied as key materials in various optical devices (Tagaya et al. 2003). However, optical polymers tend to exhibit birefringence due to the orientation of the polymer chains during processing or heat-drawn in usage, which could lose focusing and lower the performances of the optical devices (Tagaya et al. 2003; Ohkita et al. 2004a, b). Different methods have been proposed to prepare zero-birefringence polymers for optical devices, including random copolymerization and doping with birefringent crystals and so on (Tagaya et al. 2003; Ohkita et al. 2004a, b; Tagaya and Koike 2012).

Because of their high transparency and easy processing, PMMA/TOCN nanocomposites show a high potential in applications for optical devices. Thus, the birefringence of the nanocomposites is very critical. As shown in Fig. 9, the birefringence of a neat PMMA film increased at negative direction with strain, which is an inherent property of PMMA due to the asymmetry of the molecule structure. Macroscopically, the PMMA bulk material exhibits no birefringence because the molecule chains are randomly oriented and their intrinsic optical anisotropies are thus compensated. PMMA tends to exhibit birefringence with strain due to the alignment of the molecule chains during the tensile process. On the other hand, TOCNs, with 3–4 nm width and a high aspect ratio, could be aligned during the tensile process at the same time. As shown in Fig. 9, the birefringence of PMMA/TOCN nanocomposite films increased at positive direction with strain. Clearly, the addition of TOCN changed the birefringence behaviors because the signs of intrinsic birefringence of PMMA and TOCN are negative and positive, respectively (Frka-Petesic et al. 2015). Moreover, with increasing TOCN content, the birefringence of PMMA/TOCN nanocomposite increased significantly at positive direction, implying the birefringence of PMMA/TOCN nanocomposite with high weight ratio of TOCN was dominated by TOCN. Consequently, by controlling the amount of TOCN added, the birefringence of PMMA/TOCN nanocomposite can be adjusted, and even zero-birefringence can be achieved, which is very significant for various optical applications. Notably, the PMMA/TOCN nanocomposite containing 1 wt% TOCN exhibited nearly zero-birefringence (Fig. 9). A more detailed exploration of zero-birefringence of the PMMA/TOCN nanocomposite is currently undergoing in our laboratory. Rod-like organic crystals were investigated for their ability to compensate the birefringence of optical polymers to achieve zero-birefringence for optical devices. However, the synthesis of rod-like organic crystals is time consuming and needs expertize in sophisticated synthese. Our results show that TOCN could also be used to compensate the intrinsic birefringence of PMMA and achieve zero-birefringence. To the best of our knowledge, this is the first study on the ability of nanocellulose for compensation and control of the birefringence of polymer nanocomposites.

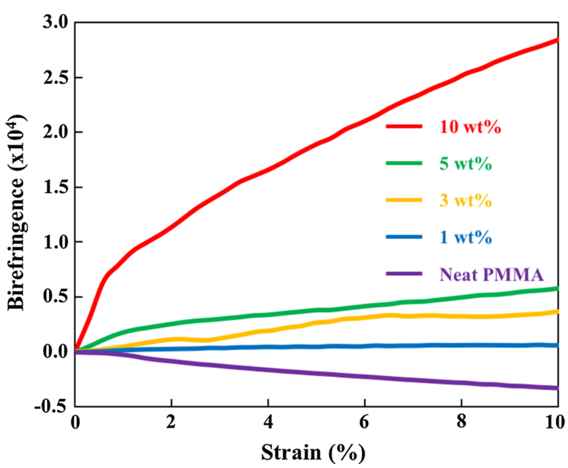

Fig. 9 Birefringence of neat PMMA and PMMA/TOCN nanocomposites (condition: $T_g + 10$ °C)

Mechanical properties

One of the important aims in the preparation of PMMA/TOCN nanocomposites is to enhance the mechanical properties, in order to broaden their engineering applications. The mechanical properties were investigated by the tensile test for neat PMMA and PMMA/TOCN nanocomposites with different contents of TOCN. Figure 10 shows that with the

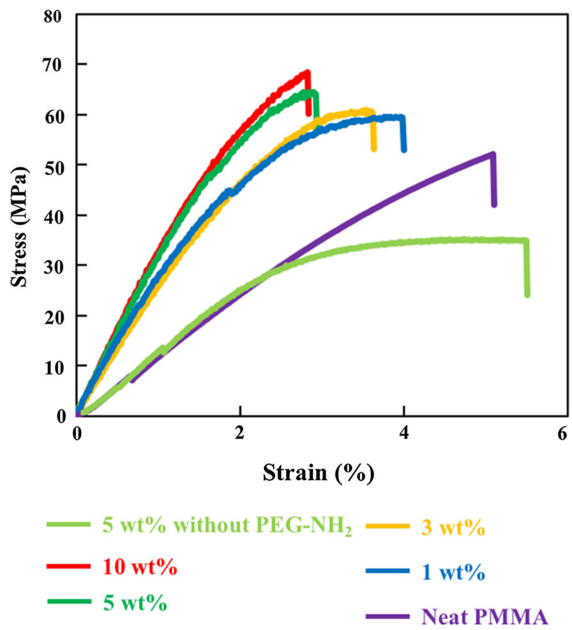

Fig. 10 Stress-strain curves of the neat PMMA, PMMA/TOCN nanocomposites with PEG-NH2 (1, 3, 5, 10 wt%) and PMMA/TOCN nanocomposites without PEG-NH2 (5 wt%)

increase of the TOCN content, the tensile strength and Young's modulus were improved, whereas elongation at break decreased (Table 1). With the addition of 1 wt% TOCN, the mechanical properties were significantly improved: the tensile strength was improved by 14.8% and Young's modulus was improved to 2.2 times, indicating the strong reinforcing ability of TOCN. On comparing neat PMMA with the PMMA/TOCN nanocomposite containing 10 wt% TOCN, it can be observed that the tensile strength was improved by 1.3-fold and Young's modulus was improved by 2.5 times. In addition, comparing the mechanical properties of PMMA/TOCN nanocomposite containing 5 wt% TOCN with/without PEG-NH_2 surface modification (Fig. 10), the PMMA/TOCN nanocomposite with surface modification showed higher tensile strength and Young's modulus, whereas the PMMA/TOCN nanocomposite without surface modification showed higher elongation, even compared with neat PMMA. As a result, the surface modification contributed to a better reinforcing performance, because of the better dispersion of TOCNs in PMMA matrix, as shown in Fig. 3.

Conclusions

In this study, PMMA/TOCN nanocomposites with high transparency and improved mechanical properties were prepared with different contents of TOCN. Owning to the surface modification with PEG-NH_2, the TOCNs were homogenously dispersed in the PMMA matrix, even with a 10 wt% content of TOCN. The optical transmittances of PMMA/TOCN nanocomposites were as high as around 90%, as a result of the good dispersion of TOCNs. Moreover, because of the opposite signs of the birefringence of PMMA and TOCN, the birefringence of PMMA/TOCN nanocomposites is controllable by adjusting the TOCN content, and even zero-birefringence can be achieved, which is significant for the application in optical devices. More importantly, the mechanical properties were significantly improved. The tensile strength had a 30% increase and Young's modulus had a 150% increase after the addition of 10 wt% TOCNs. Therefore, the PMMA/TOCN nanocomposites show high potential for the application in the engineering field and in optical devices because they exhibit with excellent transparency, thermal stability,

Table 1 Mechanical properties of the neat PMMA, PMMA/TOCN nanocomposites with PEG-NH2 (1, 3, 5, 10 wt%) and PMMA/TOCN nanocomposites without PEG-NH2 (5 wt%)

TOCN content (wt%)	Tensile strength (MPa)	Elongation at break (%)	Young's modulus (GPa)
0	52.1 ± 1.4	5.0 ± 0.2	1.1 ± 0.1
1	59.8 ± 1.2	4.0 ± 0.1	2.4 ± 0.2
3	60.8 ± 0.9	3.6 ± 0.1	2.5 ± 0.1
5	64.5 ± 2.1	3.1 ± 0.2	2.7 ± 0.2
10	68.2 ± 3.1	2.8 ± 0.3	2.8 ± 0.2
5 (without PEG-NH_2)	34.2 ± 2.4	5.4 ± 0.2	1.2 ± 0.2

adjustable birefringence and improved mechanical properties.

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